Project Insights

This customer churn analysis project shows that churn is not evenly distributed across the customer base and is strongly influenced by subscription type, plan type, geography, and support behavior. The overall churn rate was **24.3%**, while the retention rate was **75.7%**, which indicates that roughly one in four customers in the dataset was classified as churned. The average revenue per user (ARPU) was **17.51**, the average customer tenure was about **1402 days**, and the estimated revenue at risk from churned customers was **2,223,747**.

The analysis also showed clear churn differences across customer segments. The **Basic** plan had the highest churn rate at **31.61%**, followed by **Standard** at **21.88%**, and **Premium** at **12.10%**. A similar pattern appeared in subscription type analysis, where **Organic** customers churned the most, followed by **Paid**, and then **Refferal**. This suggests that lower-tier or less committed customer groups are more likely to leave the service.

Geographic analysis revealed strong variation in churn across states. Some states showed very low churn, while others showed much higher churn, with values ranging from around **5%** to nearly **40%**. This means churn behavior is not uniform and may be influenced by regional market behavior, customer expectations, or service quality differences.

The support data also added useful context. The support table contained **37,500 complaint records** from **30,000 unique customers**, which shows that not every customer complained and some customers raised multiple complaints. This made the support analysis more realistic and helped connect churn with customer service interactions.

Overall, the project highlights that churn can be better understood by combining subscription behavior, demographics, geography, and support signals. It also shows that retention strategy should focus more on lower-value plans, high-risk regions, and customers showing signs of dissatisfaction.

Project Summary

This project is a complete **Customer Churn Analysis and Customer Intelligence** workflow built with **Python, Pandas, SQLite, Matplotlib, and Seaborn**. It started by importing three Excel sheets into SQLite as separate tables, then loading them into Pandas for cleaning, transformation, analysis, and visualization. The Customerid column was used as the common key across all tables, allowing the data to be merged into a final analytical dataset.

The project included full data cleaning, such as datatype conversion, missing value handling, standardization of categorical fields, and removal of unnecessary columns.

Feature engineering was then applied to create useful business variables like Churn_flag, Customer_age, tenure_days, Churn_risk, and Cancellation_month. These transformations turned the raw dataset into a structured churn analytics model suitable for business reporting and insight generation.

After preprocessing, the merged dataframe contained **650,000 records**, and the project used both Matplotlib and Seaborn to explore churn trends, plan-level differences, country and state patterns, correlation structure, and multi-dimensional relationships. The final analysis was also supported with pivot tables to summarize churn by plan type.

This project demonstrates end-to-end data analytics capability: data ingestion, database management, cleaning, feature engineering, statistical analysis, visualization, and business insight generation. It is a practical example of how raw customer and subscription data can be converted into actionable retention strategy insights.